

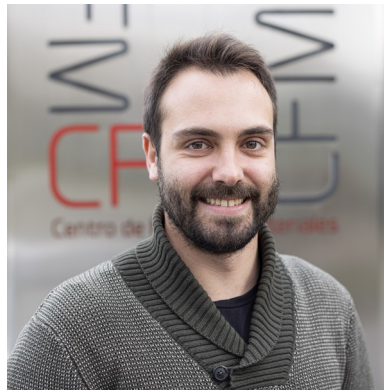
Optical spatial dispersion via Wannier interpolation

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Óscar Pozo

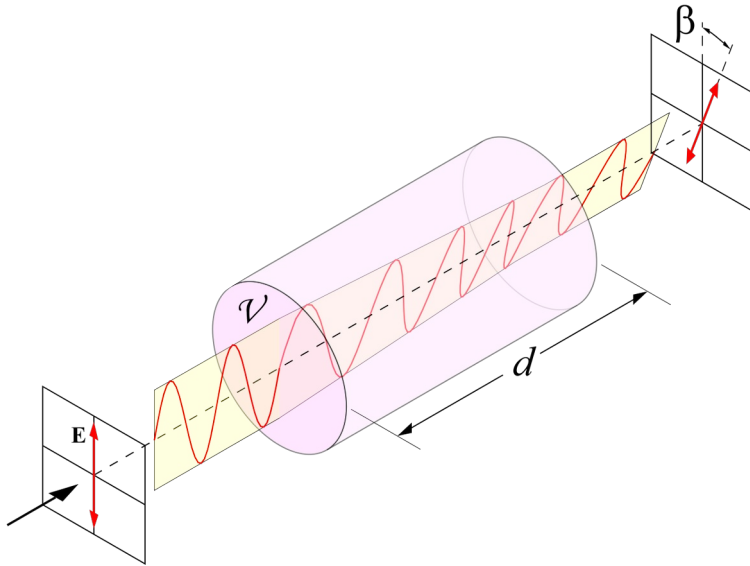


Stepan Tsirkin

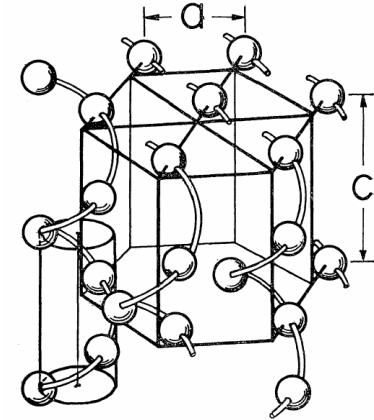


David Vanderbilt

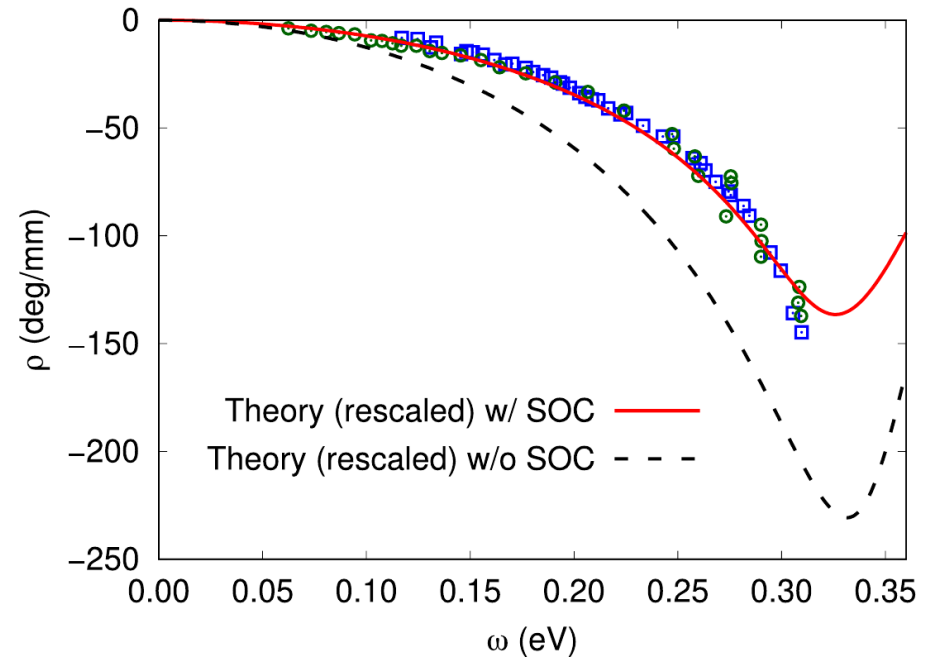
Rotatory power of trigonal tellurium



$$\text{Rotatory power: } \rho = \frac{\beta}{d}$$



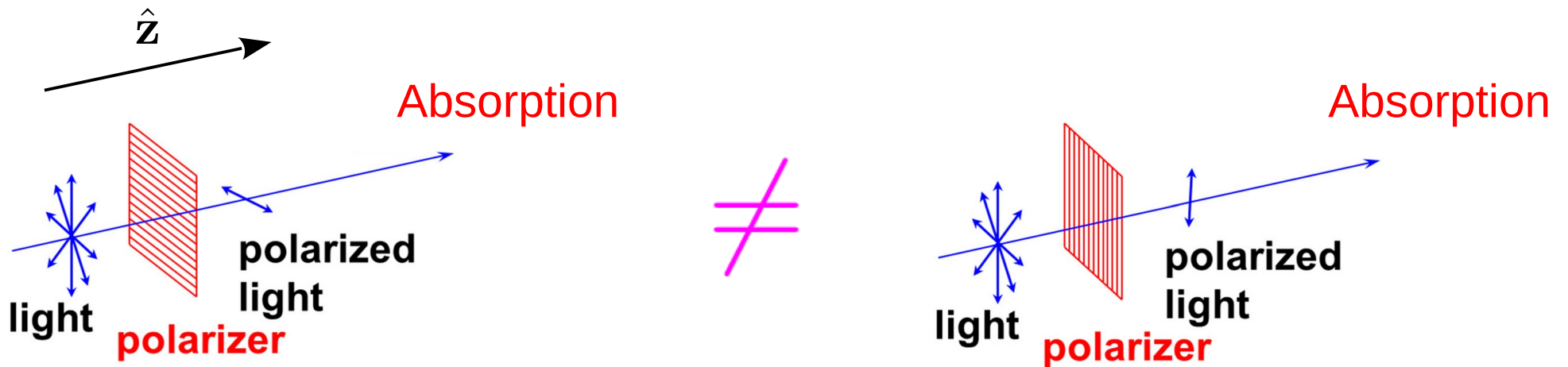
A. Urru, I. Souza, O. P. Ocaña, S. S. Tsirkin
and D. Vanderbilt
Phys. Rev. B 112, 045201 (2025)



Optical absorption

Absorption coefficient α :

$$\alpha(\mathbf{E}^\omega \parallel \hat{\mathbf{x}}) \neq \alpha(\mathbf{E}^\omega \parallel \hat{\mathbf{y}})$$



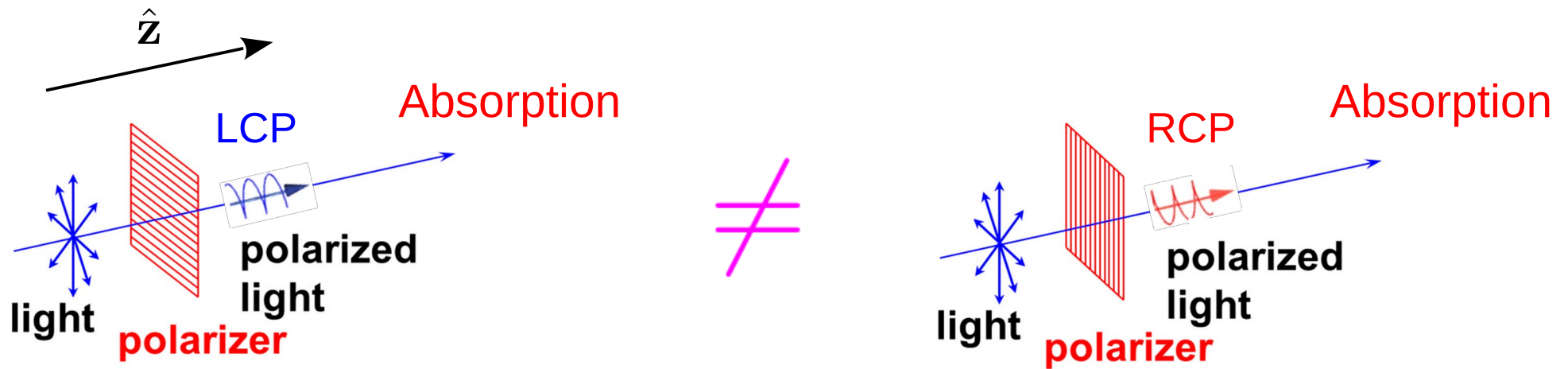
Linear dichroism (LD)

$$\varepsilon''_{xx} \neq \varepsilon''_{yy}$$

Optical absorption

Absorption coefficient α :

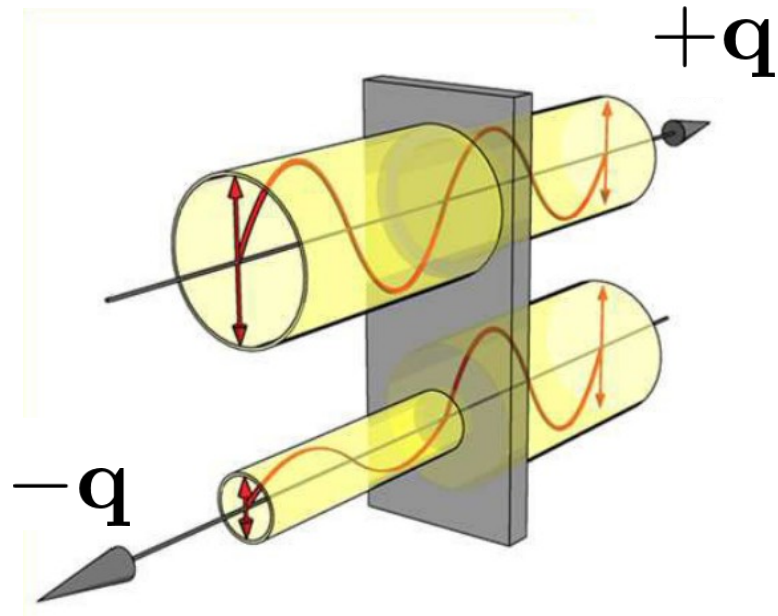
$$\alpha(\mathbf{E}^\omega \parallel \hat{\mathbf{x}} + i\hat{\mathbf{y}}) \neq \alpha(\mathbf{E}^\omega \parallel \hat{\mathbf{x}} - i\hat{\mathbf{y}})$$



Circular dichroism (CD)

$$\varepsilon'_{xy} \neq \varepsilon'_{yx}$$

Nonreciprocal directional dichroism

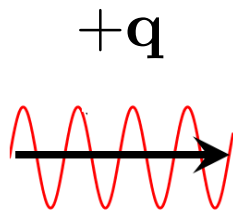


$$\varepsilon''_{yy}(+\mathbf{q}) \neq \varepsilon''_{yy}(-\mathbf{q})$$

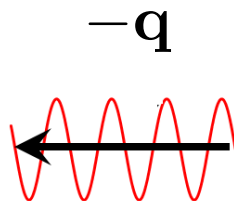
“optical diode effect”

Reciprocity and circular dichroism

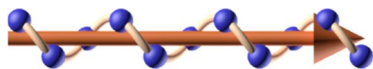
Differential absorption of LCP and RCP light



$$\alpha(\text{LCP}) > \alpha(\text{RCP})$$



$$\alpha(\text{LCP}) > \alpha(\text{RCP})$$



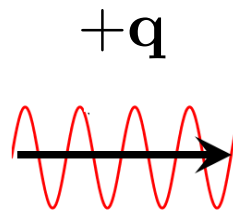
Chiral crystal

Natural CD

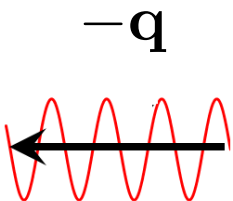
(reciprocal)

Reciprocity and circular dichroism

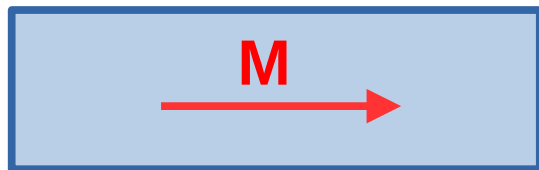
Differential absorption of LCP and RCP light



$$\alpha(\text{LCP}) > \alpha(\text{RCP})$$



$$\alpha(\text{LCP}) < \alpha(\text{RCP})$$

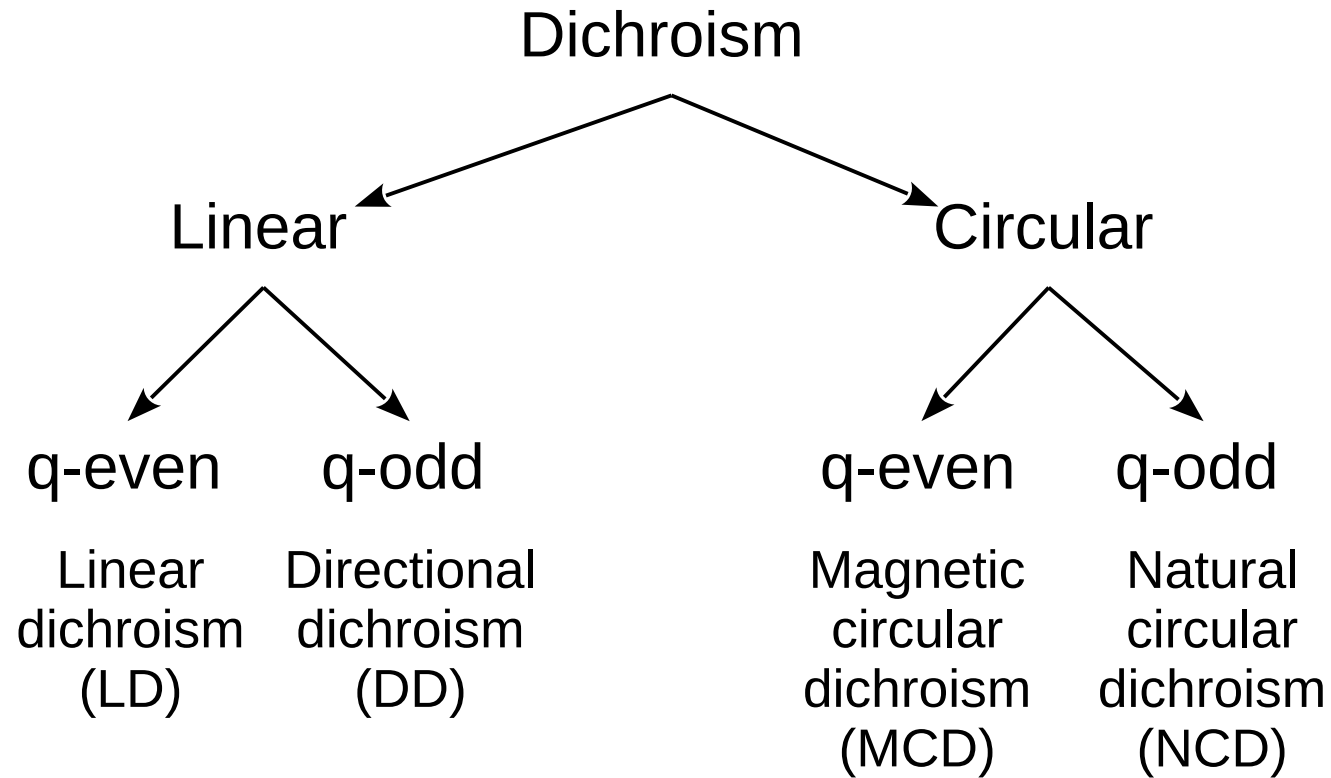


Ferromagnet

Magnetic CD

(nonreciprocal)

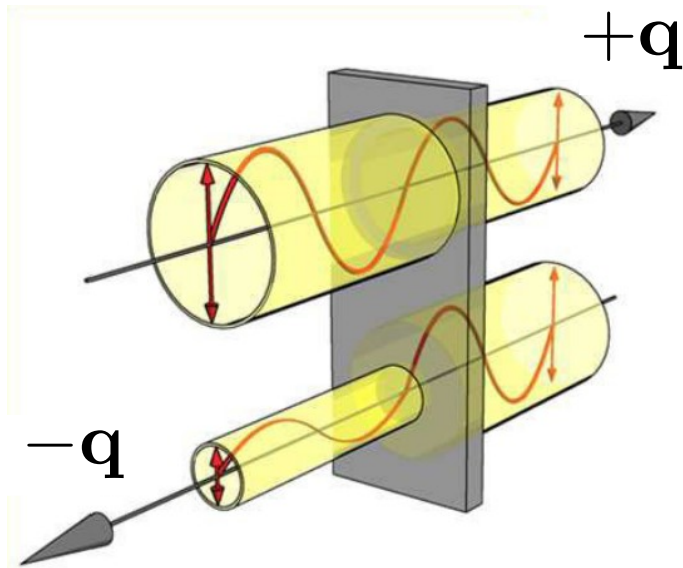
Optical absorption



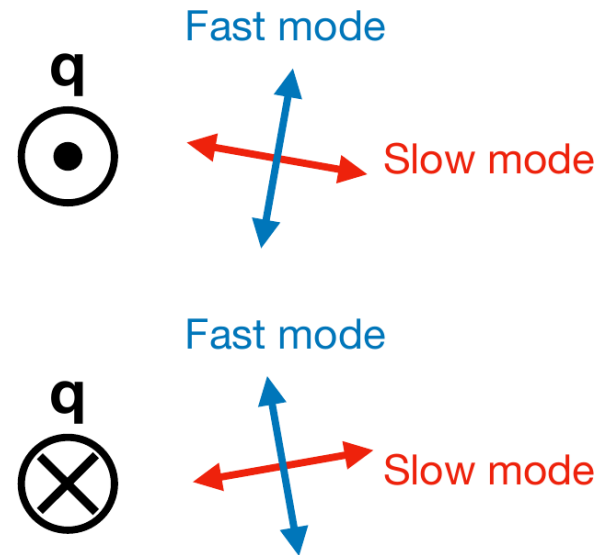
Dichroism vs birefringence

difference in
absorption

difference in
refraction



Directional dichroism

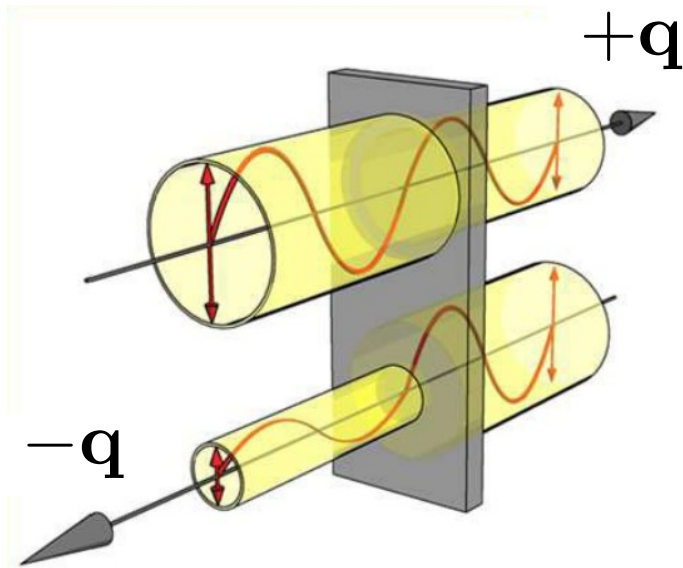


Directional birefringence

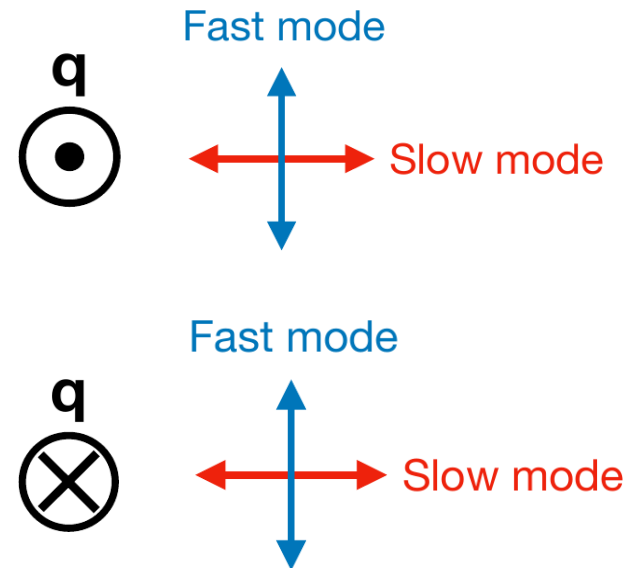
Dichroism vs birefringence

difference in
absorption

difference in
refraction

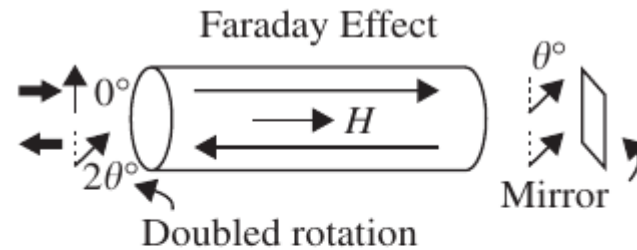
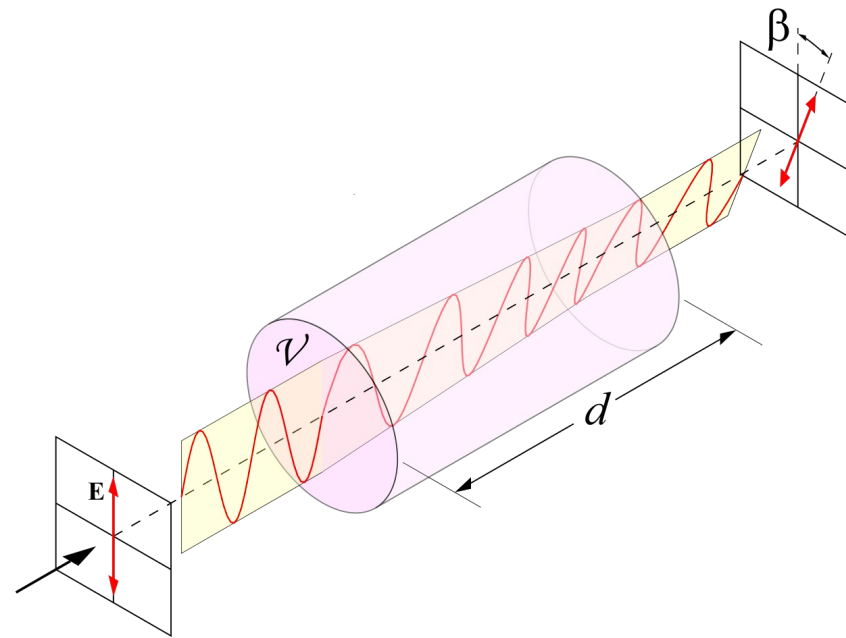


Directional dichroism

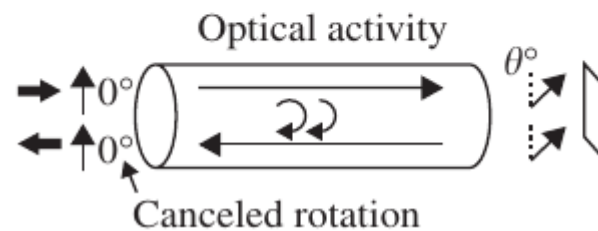


Linear birefringence

Circular birefringence



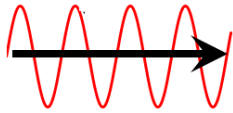
Magnetic
(nonreciprocal)



Natural
(reciprocal)

Spatial dispersion

$$e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)}$$



$$\lambda \gg a$$

$$q \ll 2\pi/a$$

Expand in powers of $\mathbf{q}$

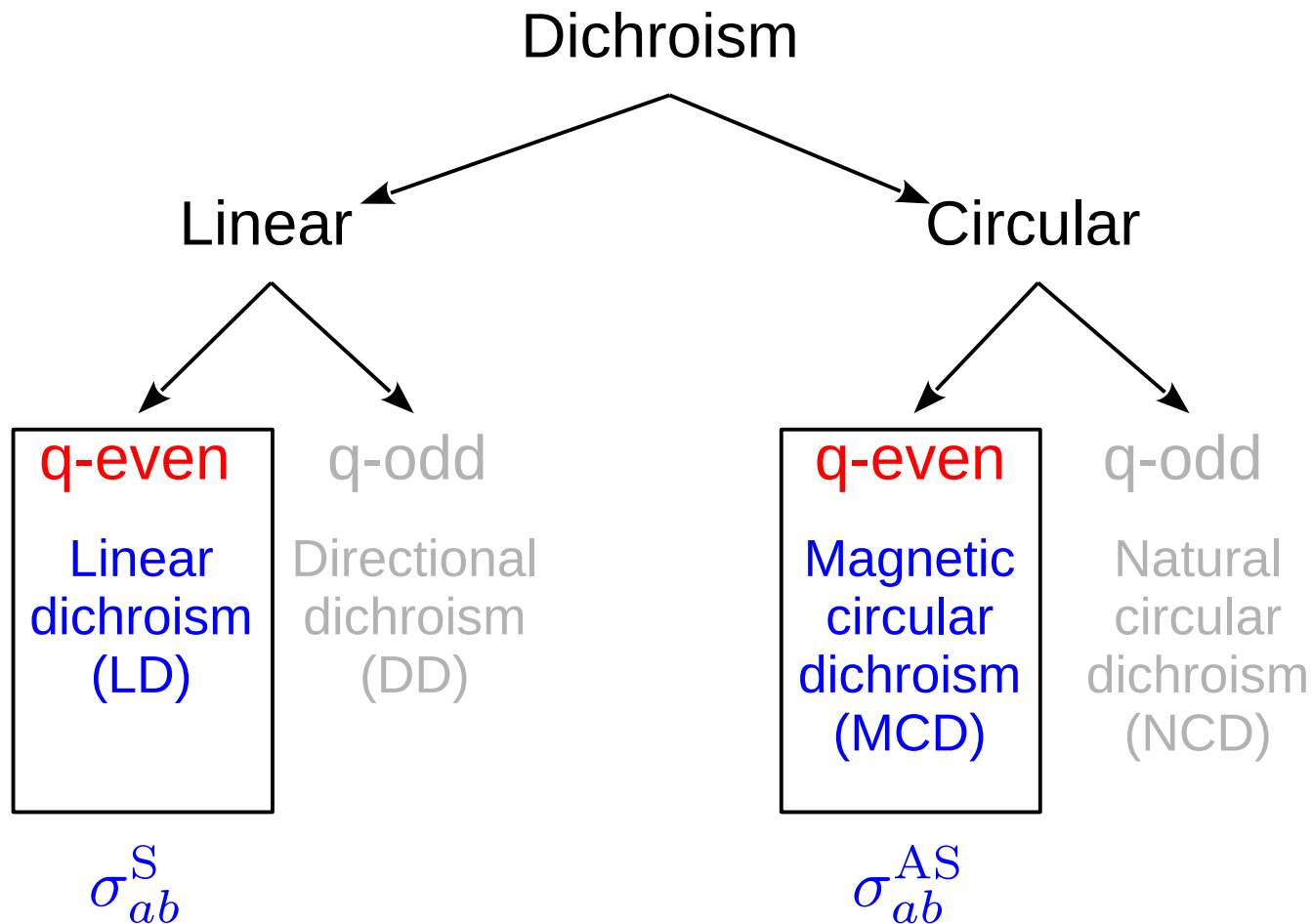
Optical conductivity: $j_a = \sigma_{ab} E_b$

$$\sigma_{ab}(\omega, \mathbf{q}) = \sigma_{ab}(\omega) + q_c \sigma_{ab,c}(\omega) + \mathcal{O}(q^2)$$

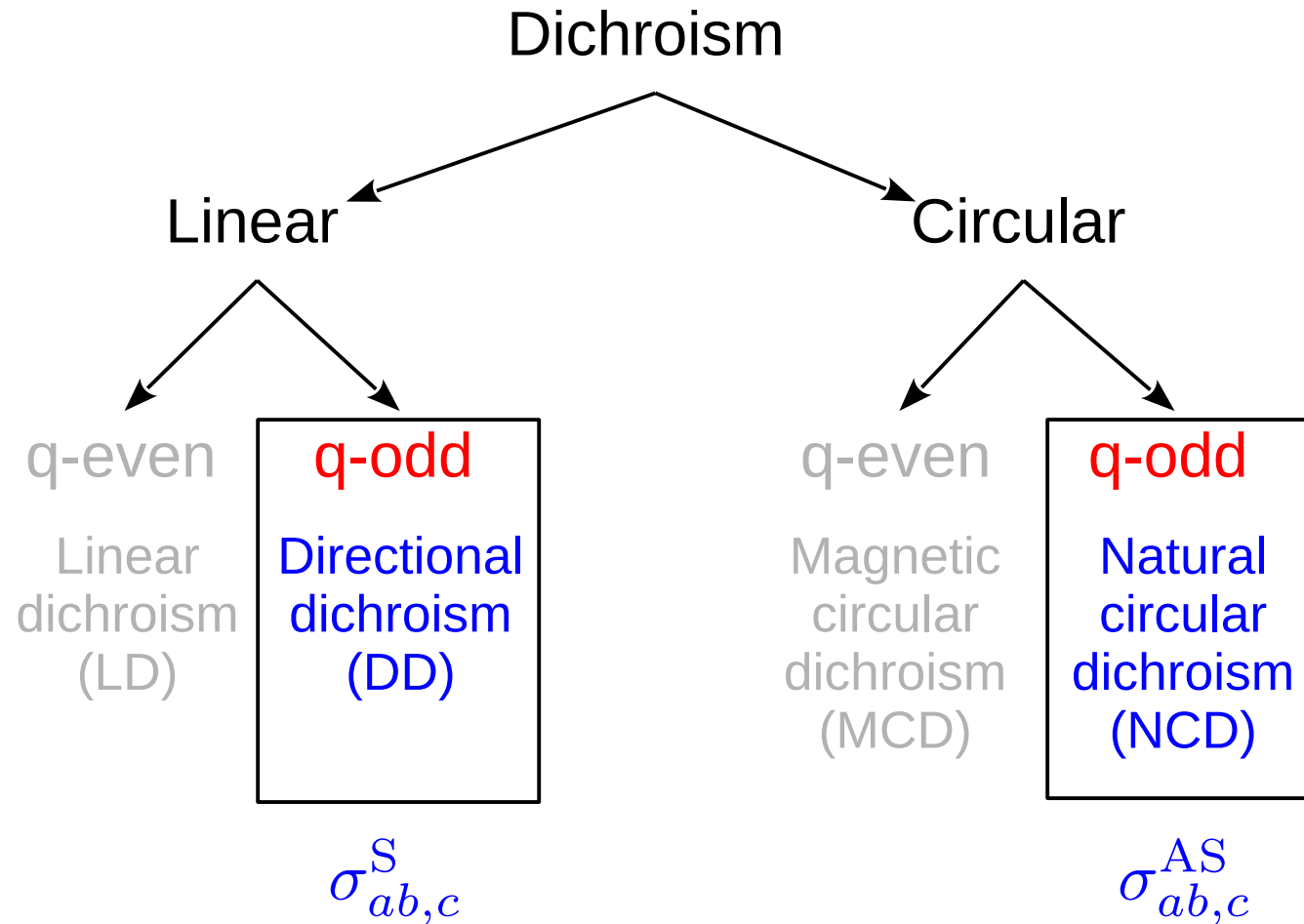
Dielectric function: $D_a = \varepsilon_{ab} E_b$

$$\varepsilon_{ab}(\omega, \mathbf{q}) = \delta_{ab} + \frac{i}{\varepsilon_0 \omega} \sigma_{ab}(\omega, \mathbf{q})$$

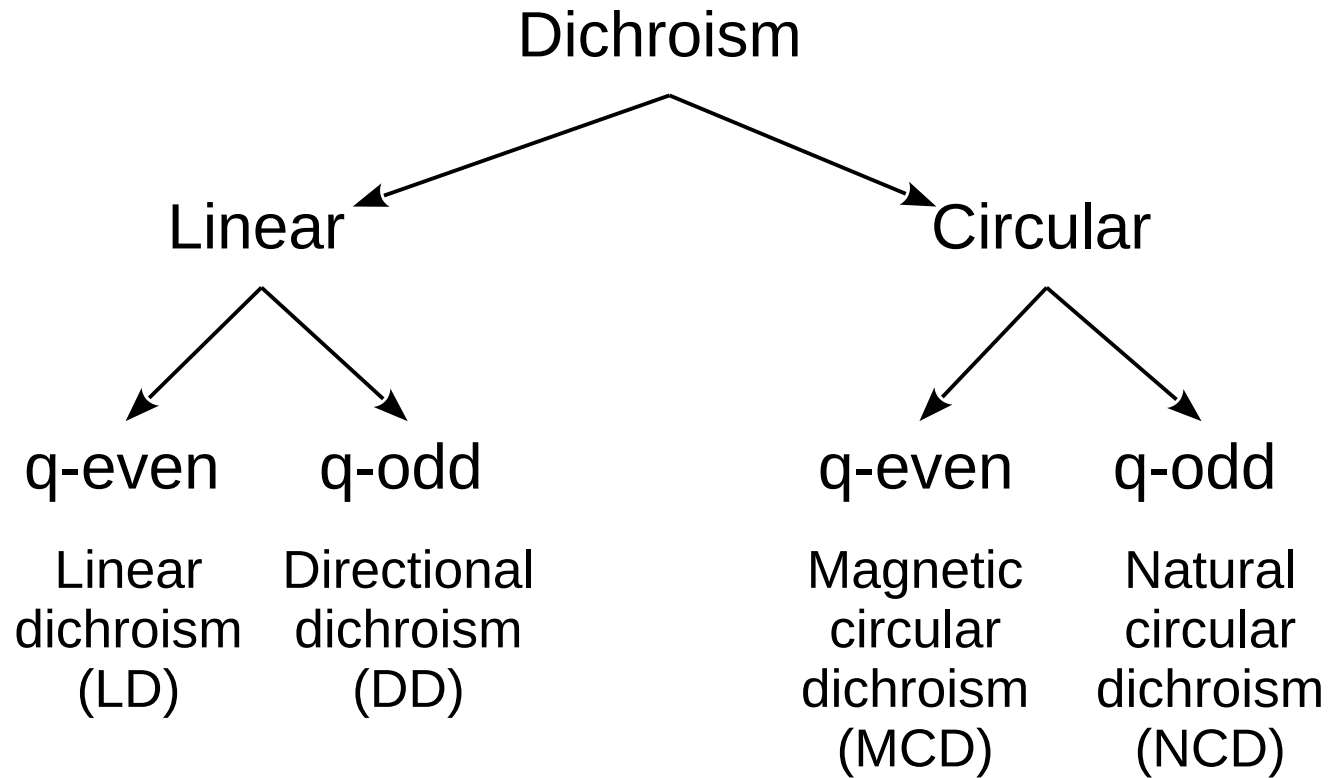
Non spatially-dispersive effects: $O(q^0)$



Spatially-dispersive effects: $O(q^1)$



Symmetry properties



$\mathcal{P}$	+	—	+	—
$\mathcal{T}$	+	—	—	+
$\mathcal{PT}$	+	+	—	—
		AFM	FM	Chiral

Ab initio scheme

Our approach:

- Start from Kubo formula for $\sigma_{ab}(\omega, \mathbf{q})$
- Expand to first order in $\mathbf{q}$ at finite ω
- Evaluate BZ integrals by “Wannier interpolation”
 - Avoids full DFT calculation at each $\mathbf{k}$
 - No band-truncation errors in dichroism

Limitations:

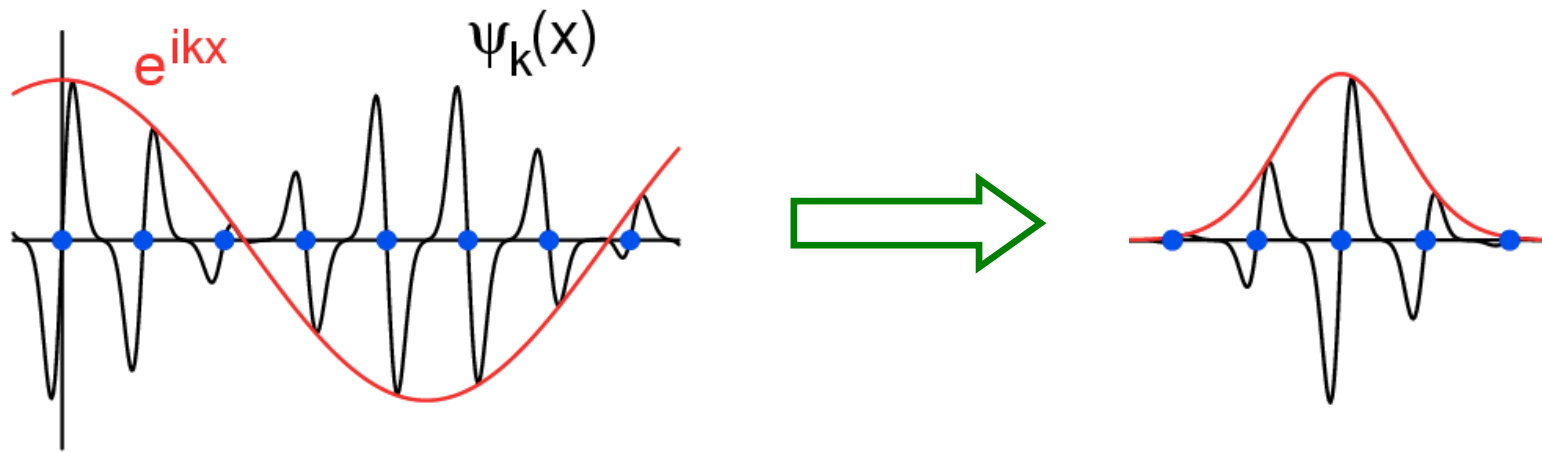
- Local-field effects not included
- Truncated Kramers-Krönig for birefringence

How to overcome: Asier Zabalo's talk

From Bloch functions to Wannier orbitals

Choose Wannier functions as

$$\langle \mathbf{r} | \mathbf{R} j \rangle = w_j(\mathbf{r} - \mathbf{R}) = \int_{\text{BZ}} \psi_{j\mathbf{k}}(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{R}} d\mathbf{k}$$



Form wave-packet = “Wannier function”

Wannier $\leftrightarrow$ “Exact” Tight-Binding

Build

$$H_{ij}^{(W)}(\mathbf{R}) = \langle \mathbf{0}i | H_{\text{KS}} | \mathbf{R}j \rangle \quad \leftarrow \text{On-site energies, hoppings}$$

Wannierized bands
(few)

Short-ranged

Fourier

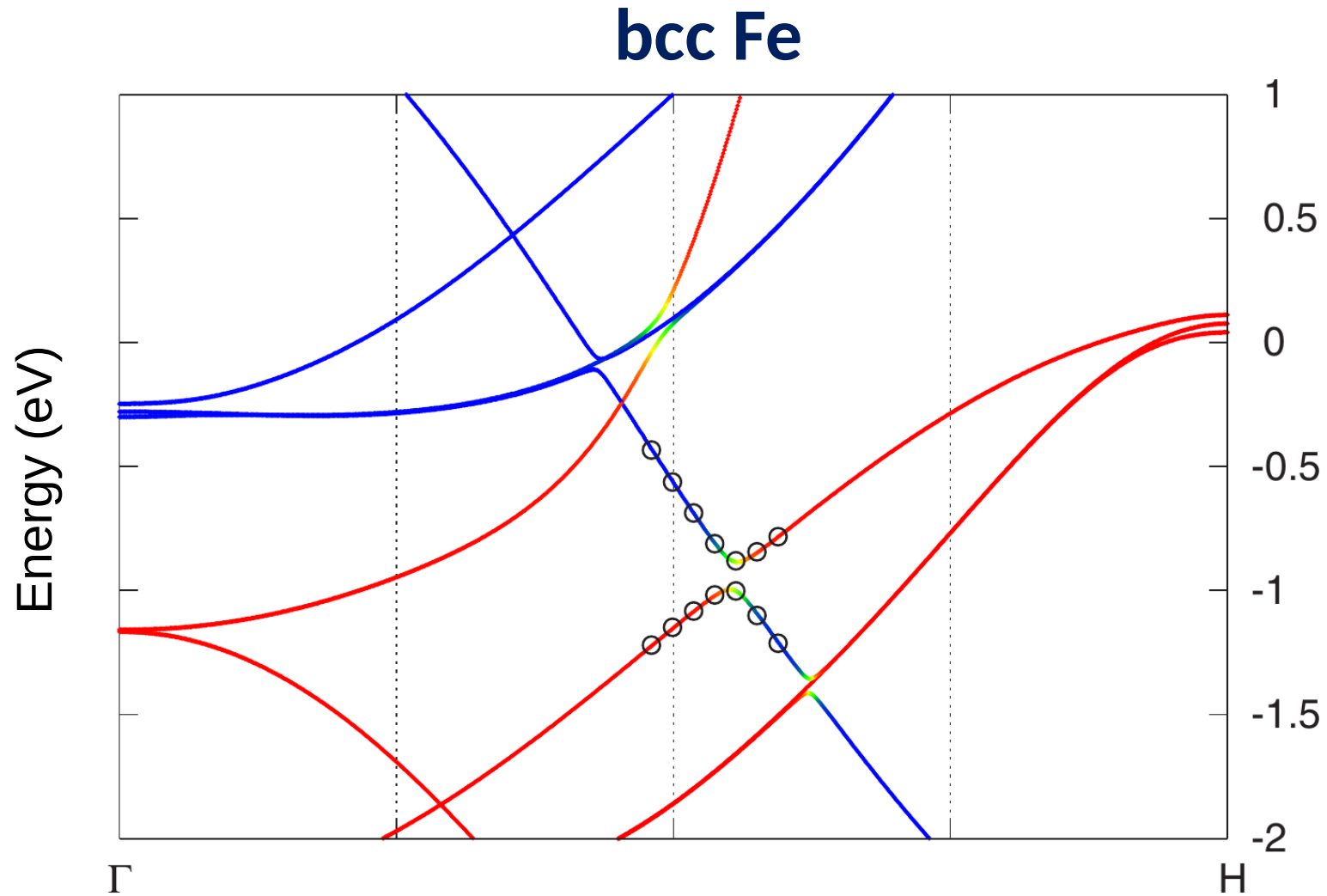
$$H_{ij}^{(W)}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} + \tau_j - \tau_i)} H_{ij}^{(W)}(\mathbf{R})$$

$\langle \mathbf{0}j | \mathbf{r} | \mathbf{0}j \rangle$

Diagonalize

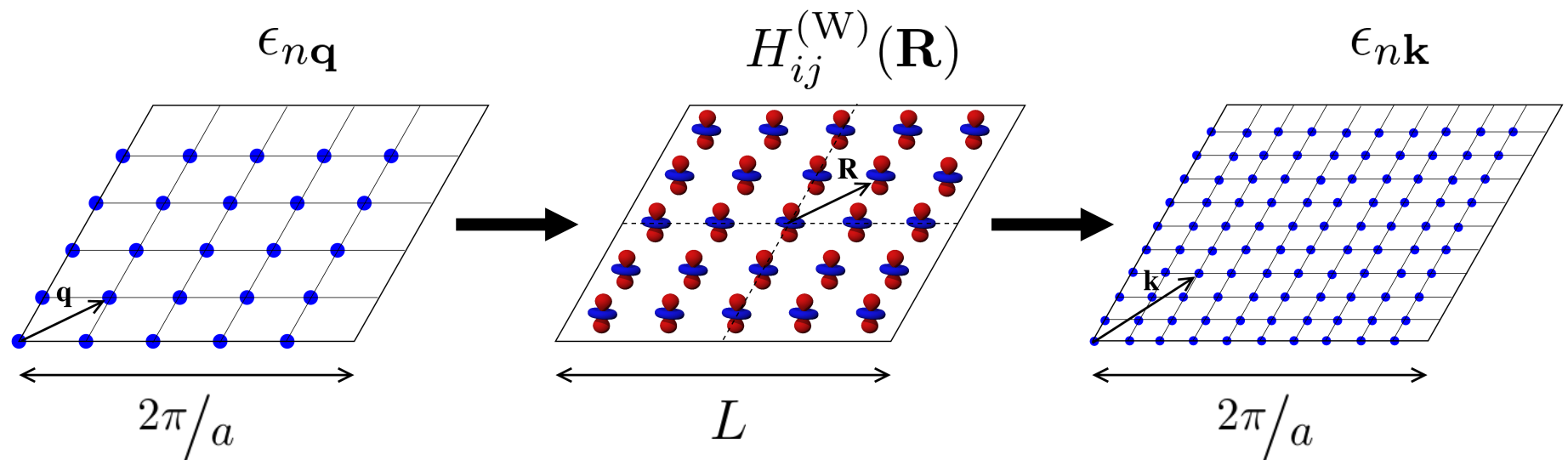
$$H_{ij}^{(W)}(\mathbf{k}) \rightarrow \{\epsilon_{n\mathbf{k}}\}$$

WFs as “perfect” interpolators



J. R. Yates, X. Wang, D. Vanderbilt and I. Souza, Phys. Rev. B 75, 195121 (2007)

Wannier interpolation



Not limited to band interpolation

- Berry curvature
- Orbital magnetization
- Optical matrix elements
- ...

Example: MCD spectrum

$$\begin{aligned} \text{Im } \sigma_{ab}^{\text{AS}}(\omega) = & -\frac{\pi e^2 \omega}{\hbar} \sum_{nl} \int_{\text{BZ}} \frac{d^3 k}{(2\pi)^3} f_n (1 - f_l) \\ & \times \text{Im} (A_{a,nl} A_{b,ln}) \delta(\omega - \omega_{ln}) \end{aligned}$$

Example: MCD spectrum

$$\begin{aligned} \text{Im } \sigma_{ab}^{\text{AS}}(\omega) = & -\frac{\pi e^2 \omega}{\hbar} \sum_{nl} \int_{\text{BZ}} \frac{d^3 k}{(2\pi)^3} f_n (1 - f_l) \\ & \times \text{Im} (A_{a,nl} A_{b,ln}) \delta(\omega - \omega_{ln}) \end{aligned}$$

$$\begin{aligned} l \neq n : \quad \mathbf{A}_{nl}(\mathbf{k}) = & i \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} | u_{l\mathbf{k}} \rangle \quad \text{interband dipole} \\ = & i \frac{\langle \psi_{n\mathbf{k}} | \hbar \mathbf{v} | \psi_{l\mathbf{k}} \rangle}{\epsilon_{l\mathbf{k}} - \epsilon_{n\mathbf{k}}} \end{aligned}$$

- f_n , f_l and ω_{ln} from band interpolation
- What about $\mathbf{A}_{nl}$?

Interpolation of A_{nl}

$$H_{ij}^{(W)}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)} \langle \mathbf{0}i | H_{\text{KS}} | \mathbf{R}j \rangle$$

$$\mathbf{A}_{ij}^{(W)}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)} \langle \mathbf{0}i | \mathbf{r} - \boldsymbol{\tau}_j | \mathbf{R}j \rangle$$

$$H^{(W)} \|\phi_{n\mathbf{k}}\rangle\rangle = \epsilon_{n\mathbf{k}} \|\phi_{n\mathbf{k}}\rangle\rangle \quad (\text{Wannier tight-binding})$$

$$\mathbf{A}_{nl}(\mathbf{k}) = i \frac{\langle\langle \phi_{n\mathbf{k}} | \boldsymbol{\nabla}_{\mathbf{k}} H^{(W)}(\mathbf{k}) | \phi_{l\mathbf{k}} \rangle\rangle}{\epsilon_{l\mathbf{k}} - \epsilon_{n\mathbf{k}}} + \langle\langle \phi_{n\mathbf{k}} | \mathbf{A}^{(W)}(\mathbf{k}) | \phi_{l\mathbf{k}} \rangle\rangle$$

“internal”

“external”

Interpolation of A_{nl}

$$H_{ij}^{(W)}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)} \langle \mathbf{0}i | H_{\text{KS}} | \mathbf{R}j \rangle$$

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Internal term resembles

$$\mathbf{A}_{nl}(\mathbf{k}) = i \frac{\langle \psi_{n\mathbf{k}} | \hbar \mathbf{v} | \psi_{l\mathbf{k}} \rangle}{\epsilon_{l\mathbf{k}} - \epsilon_{n\mathbf{k}}}$$

Interpolation of A_{nl}

$$H_{ij}^{(W)}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)} \langle \mathbf{0}i | H_{\text{KS}} | \mathbf{R}j \rangle$$

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External term is a small correction.

Interpolation of A_{nl}

$$H_{ij}^{(W)}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)} \langle \mathbf{0}i | H_{\text{KS}} | \mathbf{R}j \rangle$$

$$\mathbf{A}_{ij}^{(W)}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot (\mathbf{R} + \boldsymbol{\tau}_j - \boldsymbol{\tau}_i)} \langle \mathbf{0}i | \mathbf{r} - \boldsymbol{\tau}_j | \mathbf{R}j \rangle \quad (\text{off-diagonal})$$

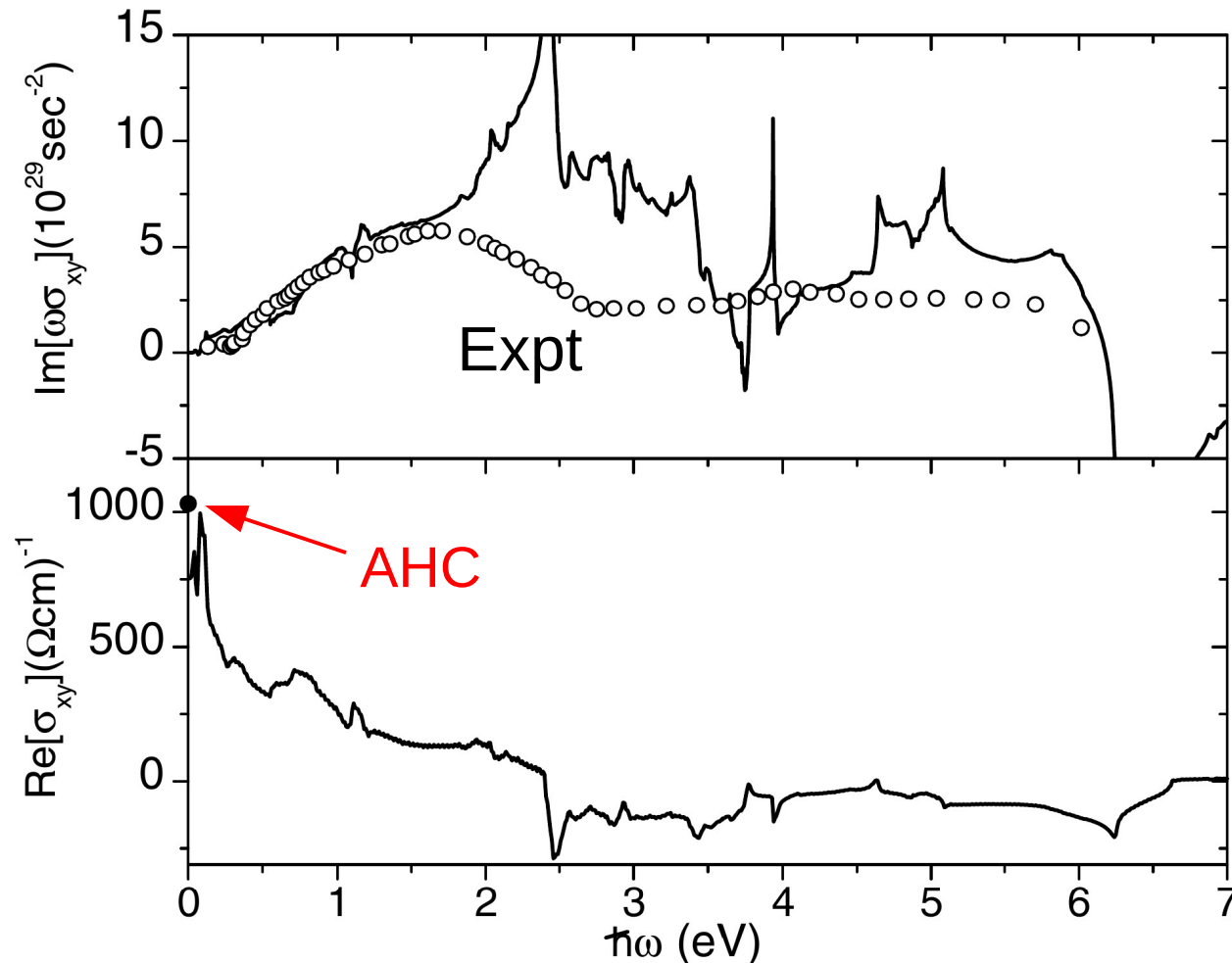
$$H^{(W)} \|\phi_{n\mathbf{k}}\rangle\rangle = \epsilon_{n\mathbf{k}} \|\phi_{n\mathbf{k}}\rangle\rangle \quad (\text{Wannier tight-binding})$$

$$\mathbf{A}_{nl}(\mathbf{k}) = i \frac{\langle\langle \phi_{n\mathbf{k}} | \boldsymbol{\nabla}_{\mathbf{k}} H^{(W)}(\mathbf{k}) | \phi_{l\mathbf{k}} \rangle\rangle}{\epsilon_{l\mathbf{k}} - \epsilon_{n\mathbf{k}}} + \langle\langle \phi_{n\mathbf{k}} | \mathbf{A}^{(W)}(\mathbf{k}) | \phi_{l\mathbf{k}} \rangle\rangle$$

External term is a small correction. It vanishes if one takes

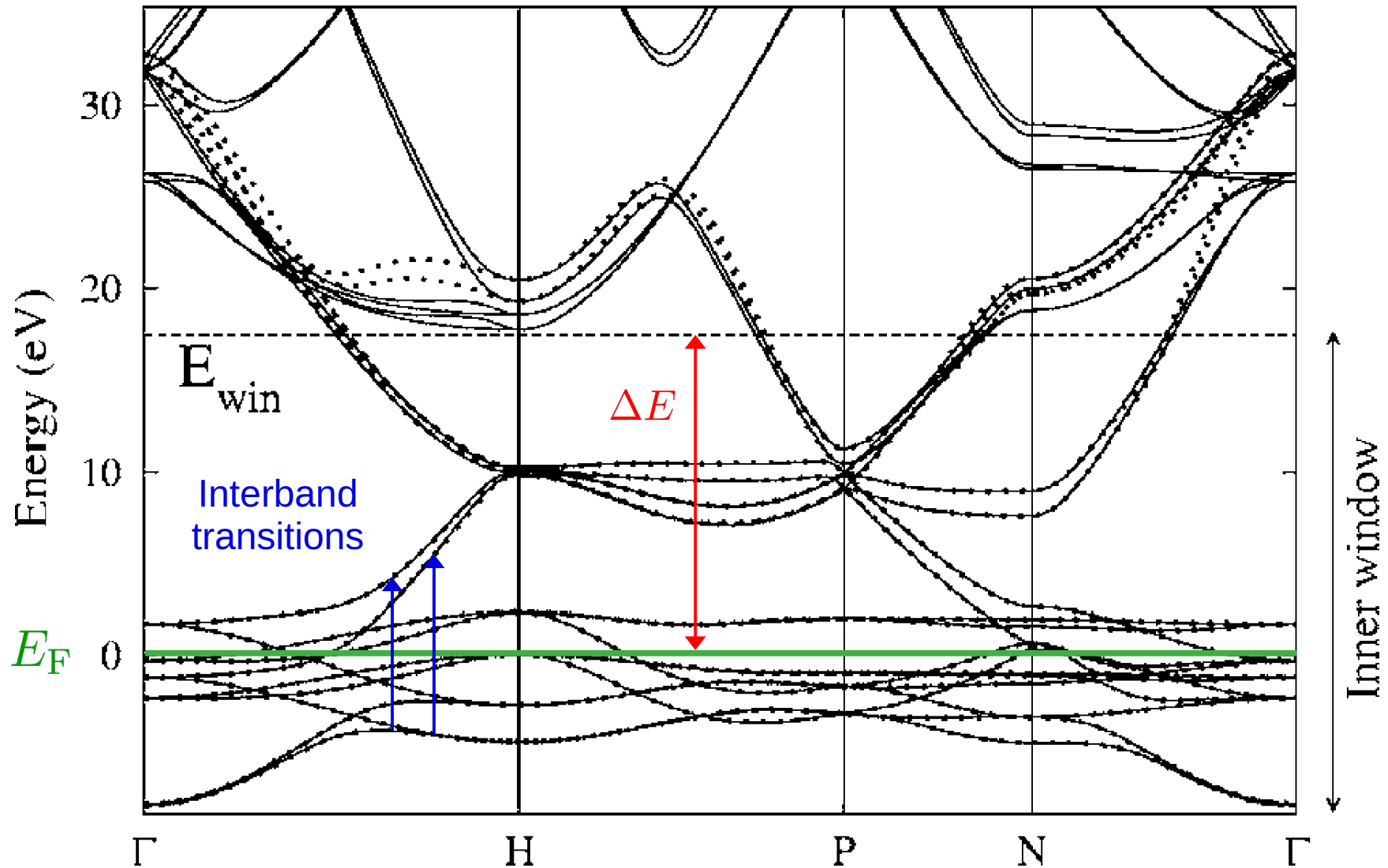
$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle \simeq \boldsymbol{\tau}_i \delta_{ij} \delta_{\mathbf{R},0} \quad (\text{diagonal})$$

MCD spectrum of bcc Fe



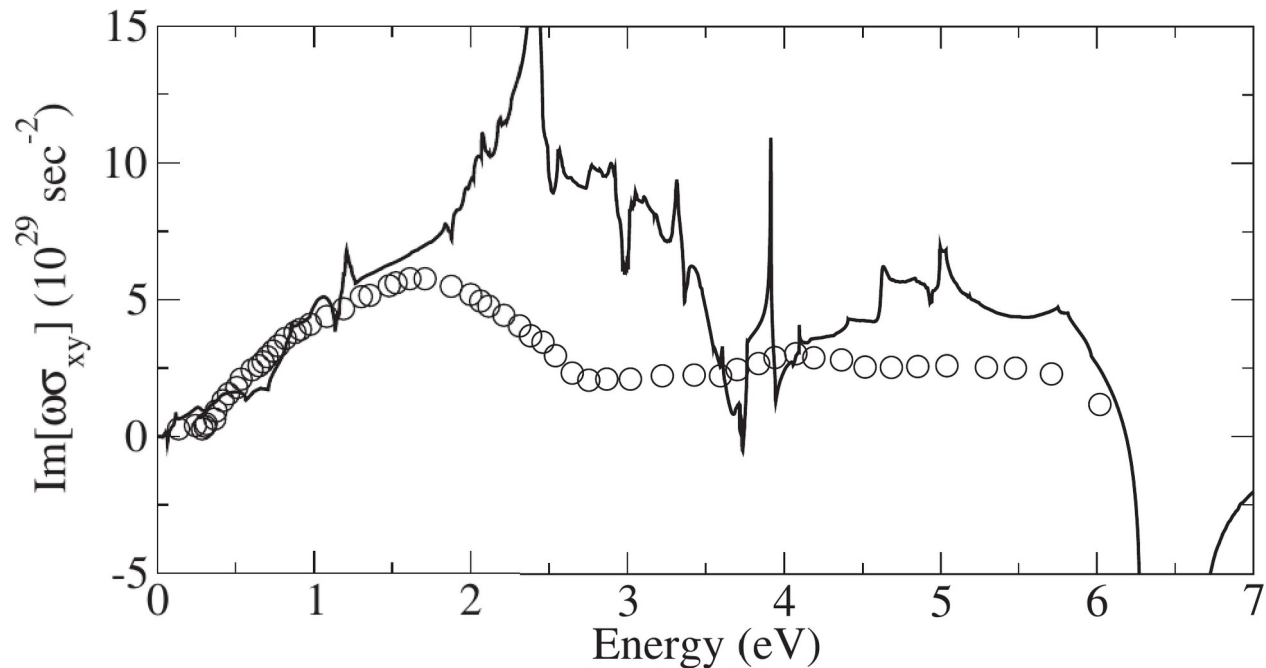
- Dense **k**-point sampling
- Each **k** took a **full** *ab initio* calculation
- Top: sum over states
- Bottom: Kramers-Kronig

Interpolation: Energy windows



$$\text{win_froz_max} = E_F + \Delta E$$

Interpolated MCD spectrum



- *Ab initio* calculation on a coarse **k**-point mesh
- Interpolate to dense mesh (**inexpensive**)
- Excellent agreement with full-blown *ab initio*

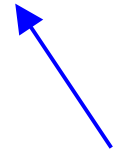
Optical spatial dispersion

$$j_a = \sigma_{ab}E_b + \sigma_{ab,c}E_bq_c + \mathcal{O}(q^2)$$

Optical spatial dispersion

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

Optical matrix elements: $\mathbf{E} \cdot \mathbf{E}$



$\text{Im} (A_{a,nl} A_{b,ln})$ in MCD

Optical spatial dispersion

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

Optical matrix elements: $\text{E1} \cdot \text{E1}$ $\text{E1} \cdot \text{M1}$
 $\text{E1} \cdot \text{E2}$

E1: electric dipole $\mathbf{d}_{nl} = -e \langle \psi_n | \mathbf{r} | \psi_l \rangle$

M1: magnetic dipole $\mathbf{m}_{nl} = -\frac{e}{2} \langle \psi_n | \mathbf{r} \times \mathbf{v} | \psi_l \rangle - \frac{eg_s}{2m_e} \langle \psi_n | \mathbf{S} | \psi_l \rangle$

E2: electric quadrupole $q_{nl}^{ab} = -e \langle \psi_n | r_a r_b | \psi_l \rangle$

Molecular theory

E. U. Condon, Rev. Mod. Phys. 9, 432 (1937)

Optical spatial dispersion

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

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E2: electric quadrupole $q_{nl}^{ab} = -e \langle \psi_n | r_a r_b | \psi_l \rangle$

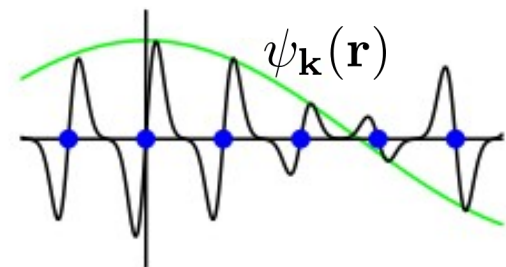
Apply to crystalline solids?

Problem: How to define $\langle \psi_{n\mathbf{k}} | \mathbf{r} | \psi_{l\mathbf{k}} \rangle$

$$\langle \psi_{n\mathbf{k}} | \mathbf{r} \times \mathbf{v} | \psi_{l\mathbf{k}} \rangle$$

$$\langle \psi_{n\mathbf{k}} | r_a r_b | \psi_{l\mathbf{k}} \rangle$$

?



Optical spatial dispersion: crystals

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

Optical matrix elements:

$\mathbf{E1} \cdot \mathbf{E1}$	$\mathbf{E1} \cdot \mathbf{M1}$
	$\mathbf{E1} \cdot \mathbf{E2}$
	$\mathbf{E1} \cdot \mathbf{v}$ (!)

Optical spatial dispersion: crystals

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

Optical matrix elements: $E1 \cdot E1$ $E1 \cdot M1$
 $E1 \cdot E2$

$E1 \cdot v \leftarrow$ band dispersion $v = \nabla_{\mathbf{k}} \epsilon$

vanishes for molecules on a supercell

origin: non-vertical transitions

K. Natori, J. Phys. Soc. Japan 39, 1013 (1975)

**A. Malashevich and I. Souza,
Phys. Rev. B 82, 245118 (2010)**

CD of carbon nanotubes

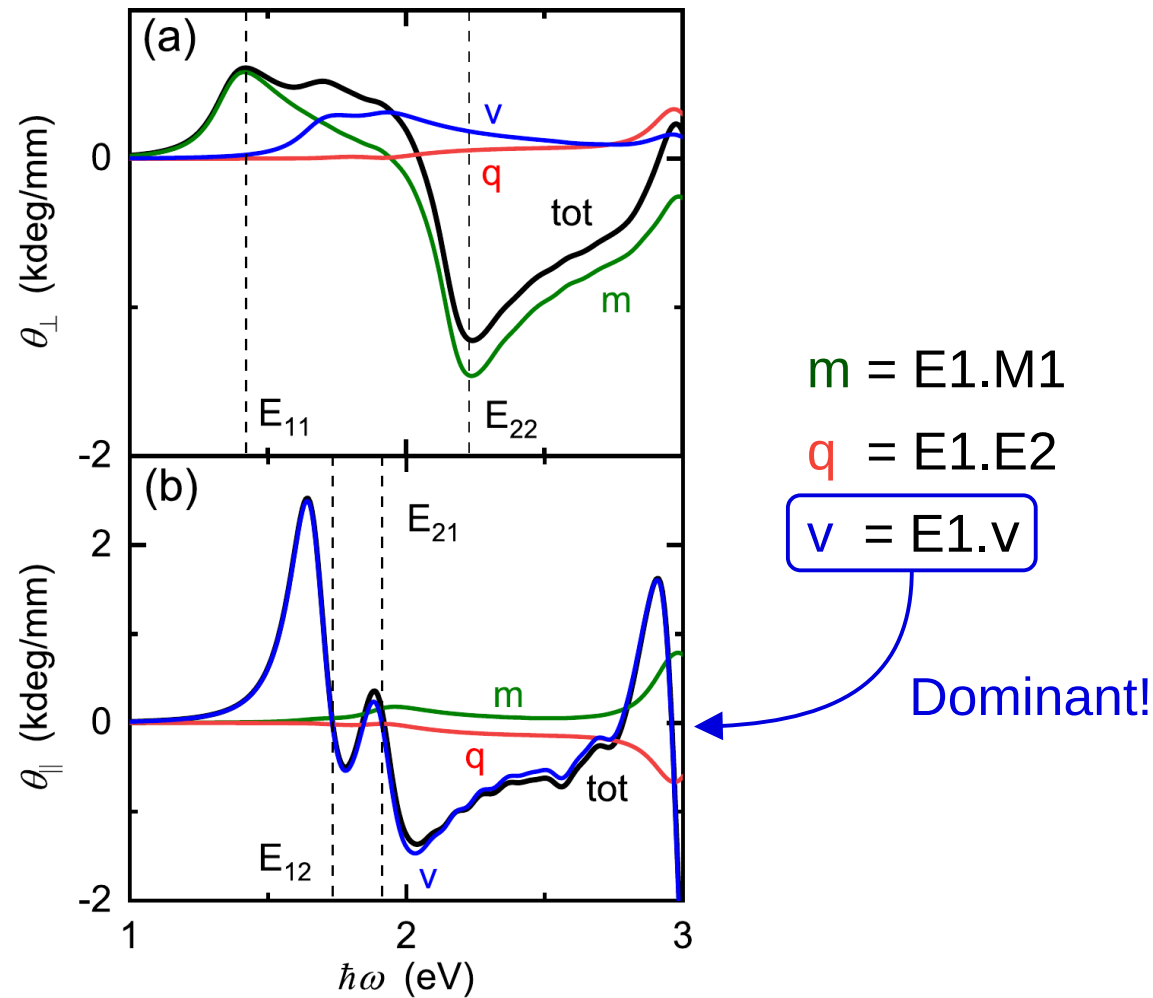


FIG. 6. CD of (6,4) CNT for light propagating (a) perpendicular to ($\theta_{\perp}$) and (b) along ($\theta_{\parallel}$) the tube axis.

Optical spatial dispersion: crystals

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

Optical matrix elements:

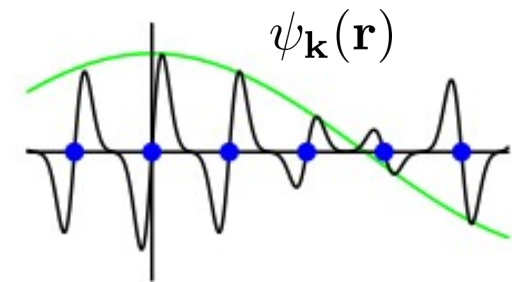
$\mathbf{E} \cdot \mathbf{E}$	$\mathbf{E} \cdot \mathbf{M}$
	$\mathbf{E} \cdot \mathbf{E}$
	$\mathbf{E} \cdot \mathbf{v}$

Problem: How to define $\langle \psi_{n\mathbf{k}} | \mathbf{r} | \psi_{l\mathbf{k}} \rangle$

$$\langle \psi_{n\mathbf{k}} | \mathbf{r} \times \mathbf{v} | \psi_{l\mathbf{k}} \rangle$$

$$\langle \psi_{n\mathbf{k}} | r_a r_b | \psi_{l\mathbf{k}} \rangle$$

?



Optical spatial dispersion: crystals

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

Optical matrix elements:

- $\text{E1} \cdot \text{E1}$
- $\text{E1} \cdot \text{M1}$
- $\text{E1} \cdot \text{E2}$
- $\text{E1} \cdot \mathbf{v}$

$(1 - |u_{l\mathbf{k}}\rangle\langle u_{l\mathbf{k}}|) |\nabla_{\mathbf{k}} u_{l\mathbf{k}}\rangle$ covariant derivative



E1: electric dipole $\mathbf{d}_{nl}(\mathbf{k}) = -ie \langle u_{n\mathbf{k}} | \mathbf{D}_{\mathbf{k}} u_{l\mathbf{k}} \rangle$

M1: magnetic dipole $\mathbf{m}_{nl}(\mathbf{k}) = \frac{e}{2i\hbar} \langle \mathbf{D}_{\mathbf{k}} u_{n\mathbf{k}} | \times \left(H_{\mathbf{k}} - \frac{\epsilon_{n\mathbf{k}} + \epsilon_{l\mathbf{k}}}{2} \right) | \mathbf{D}_{\mathbf{k}} u_{l\mathbf{k}} \rangle - \frac{eg_s}{2m_e} \langle \psi_{n\mathbf{k}} | \mathbf{S} | \psi_{l\mathbf{k}} \rangle$

E2: electric quadrupole $q_{nl}^{ab}(\mathbf{k}) = -\frac{e}{2} (\langle D_a u_{n\mathbf{k}} | D_b u_{l\mathbf{k}} \rangle + \langle D_b u_{n\mathbf{k}} | D_a u_{l\mathbf{k}} \rangle)$

Optical spatial dispersion: crystals

$$j_a = \sigma_{ab} E_b + \sigma_{ab,c} E_b q_c + \mathcal{O}(q^2)$$

Optical matrix elements:

E1 · E1	E1 · M1
	E1 · E2
	E1 · v

Set $l = n$:

E1: electric dipole $\mathbf{d}_{nn}(\mathbf{k}) = 0$

M1: magnetic dipole $\mathbf{m}_{nn}(\mathbf{k}) = \frac{e}{2i\hbar} \langle \nabla_{\mathbf{k}} u_{n\mathbf{k}} | \times (H_{\mathbf{k}} - \epsilon_{n\mathbf{k}}) | \nabla_{\mathbf{k}} u_{n\mathbf{k}} \rangle - \frac{eg_s}{2m_e} \langle \psi_{n\mathbf{k}} | \mathbf{S} | \psi_{n\mathbf{k}} \rangle$

E2: electric quadrupole $q_{nn}^{ab}(\mathbf{k}) = -e (\text{Re} \langle \nabla_a u_{n\mathbf{k}} | \nabla_b u_{n\mathbf{k}} \rangle - A_{a,n\mathbf{k}} A_{b,n\mathbf{k}})$

orbital moment



quantum metric



Intraband spatial dispersion
(conductors)

Optical matrix elements

k-derivative form:

$$\mathbf{d}_{nl}(\mathbf{k}) = -ie \langle u_{n\mathbf{k}} | \mathbf{D}_{\mathbf{k}} u_{l\mathbf{k}} \rangle$$

$$\mathbf{m}_{nl}^{\text{orb}}(\mathbf{k}) = \frac{e}{2i\hbar} \langle \mathbf{D}_{\mathbf{k}} u_{n\mathbf{k}} | \times \left(H_{\mathbf{k}} - \frac{\epsilon_{n\mathbf{k}} + \epsilon_{l\mathbf{k}}}{2} \right) | \mathbf{D}_{\mathbf{k}} u_{l\mathbf{k}} \rangle$$

$$q_{nl}^{ab}(\mathbf{k}) = -\frac{e}{2} (\langle D_a u_{n\mathbf{k}} | D_b u_{l\mathbf{k}} \rangle + \langle D_b u_{n\mathbf{k}} | D_a u_{l\mathbf{k}} \rangle)$$

Optical matrix elements

Sum-over-states form:

$$\mathbf{d}_{nl}(\mathbf{k}) = \begin{cases} ie \frac{\mathbf{v}_{nl}}{\omega_{nl}} & \text{if } l \neq n \\ 0 & \text{if } l = n \end{cases} \quad \text{Interband dipole}$$

intermediate states p


$$\mathbf{m}_{nl}^{\text{orb}}(\mathbf{k}) = \frac{e}{4i} \sum_{p \neq l, n} \left(\frac{1}{\omega_{pn}} + \frac{1}{\omega_{pl}} \right) \mathbf{v}_{np} \times \mathbf{v}_{pl}$$


$$q_{nl}^{ab}(\mathbf{k}) = -\frac{e}{2} \sum_{p \neq n, l} \left[\frac{v_{np}^a v_{pl}^b}{\omega_{pn} \omega_{pl}} + (a \leftrightarrow b) \right]$$

Optical matrix elements

- X. Wang and Y. Yan, Phys. Rev. B 107, 045201 (2023)

Truncated sum over intermediate states in M1, E2 transition moments

- **Wannier interpolation** starts from k-derivative form to obtain
 - “Internal” terms that resemble sum-over states
 - Additional “external” terms (not small for M1 and E2)

N.B.: Once *both* terms are included, there is *no truncation error*, even though sums are restricted to wannierized bands

⇒ truncation-free CD and DD spectra

Wannier matrix elements

Energy bands

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

Wannier matrix elements

Energy bands

$$\sigma_{ab}(\omega)$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

Wannier matrix elements

Energy bands

$$\sigma_{ab}(\omega)$$

$$\mathbf{M}_{\text{orb}}$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a H r_b | \mathbf{R}j \rangle$$

M. G. Lopez, D. Vanderbilt, T. Thonhauser and I. Souza
Phys. Rev. B 85, 014435 (2012)

Wannier matrix elements

Energy bands

$$\sigma_{ab}(\omega)$$

$$\mathbf{M}_{\text{orb}}$$

$$\sigma_{ab,c}(\omega)$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a H r_b | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a H r_b | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a r_b | \mathbf{R}j \rangle$$

A. Urru, I. Souza, O. P. Ocaña, S. S. Tsirkin and D. Vanderbilt
Phys. Rev. B 112, 045201 (2025)

Wannier matrix elements

Energy bands

$$\sigma_{ab}(\omega)$$

$$\mathbf{M}_{\text{orb}}$$

$$\sigma_{ab,c}(\omega)$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

E1

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a H r_b | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a H r_b | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a r_b | \mathbf{R}j \rangle$$

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Wannier matrix elements

Energy bands	$\sigma_{ab}(\omega)$	$\mathbf{M}_{\text{orb}}$	$\sigma_{ab,c}(\omega)$
$\langle \mathbf{0}i H \mathbf{R}j \rangle$	$\langle \mathbf{0}i H \mathbf{R}j \rangle$	$\langle \mathbf{0}i H \mathbf{R}j \rangle$	$\langle \mathbf{0}i H \mathbf{R}j \rangle$
	$\langle \mathbf{0}i \mathbf{r} \mathbf{R}j \rangle$	$\langle \mathbf{0}i \mathbf{r} \mathbf{R}j \rangle$	$\langle \mathbf{0}i \mathbf{r} \mathbf{R}j \rangle$
		M1 $\langle \mathbf{0}i r_a H r_b \mathbf{R}j \rangle$	$\langle \mathbf{0}i r_a H r_b \mathbf{R}j \rangle$
			$\langle \mathbf{0}i r_a r_b \mathbf{R}j \rangle$

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Wannier matrix elements

Energy bands

$$\sigma_{ab}(\omega)$$

$$\mathbf{M}_{\text{orb}}$$

$$\sigma_{ab,c}(\omega)$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | H | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | \mathbf{r} | \mathbf{R}j \rangle$$

$$\langle \mathbf{0}i | r_a H r_b | \mathbf{R}j \rangle$$

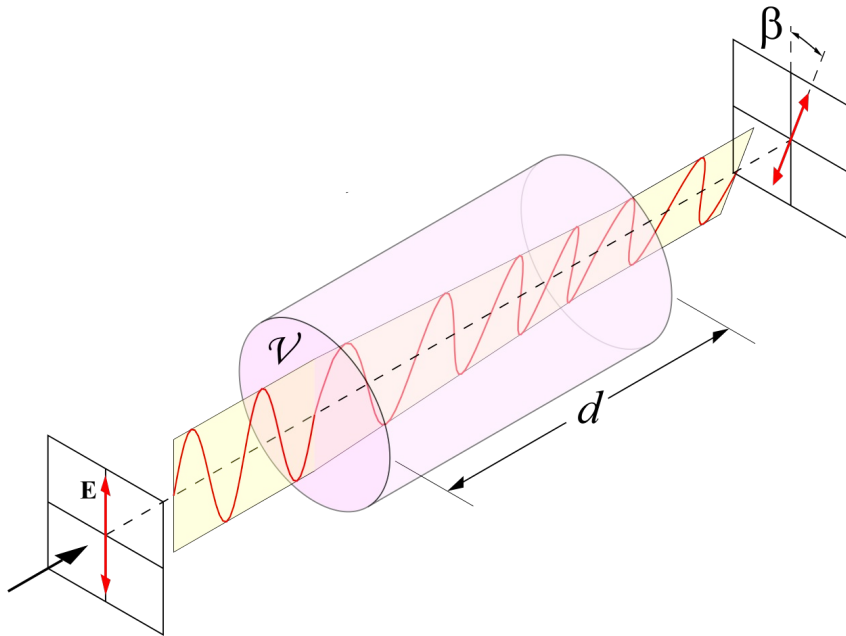
$$\langle \mathbf{0}i | r_a H r_b | \mathbf{R}j \rangle$$

E2

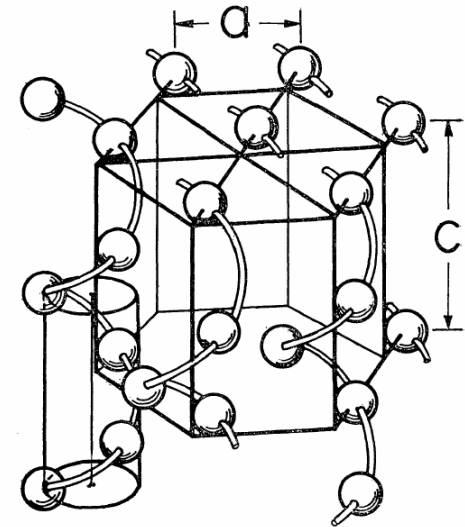
$$\langle \mathbf{0}i | r_a r_b | \mathbf{R}j \rangle$$

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Rotatory power of trigonal tellurium

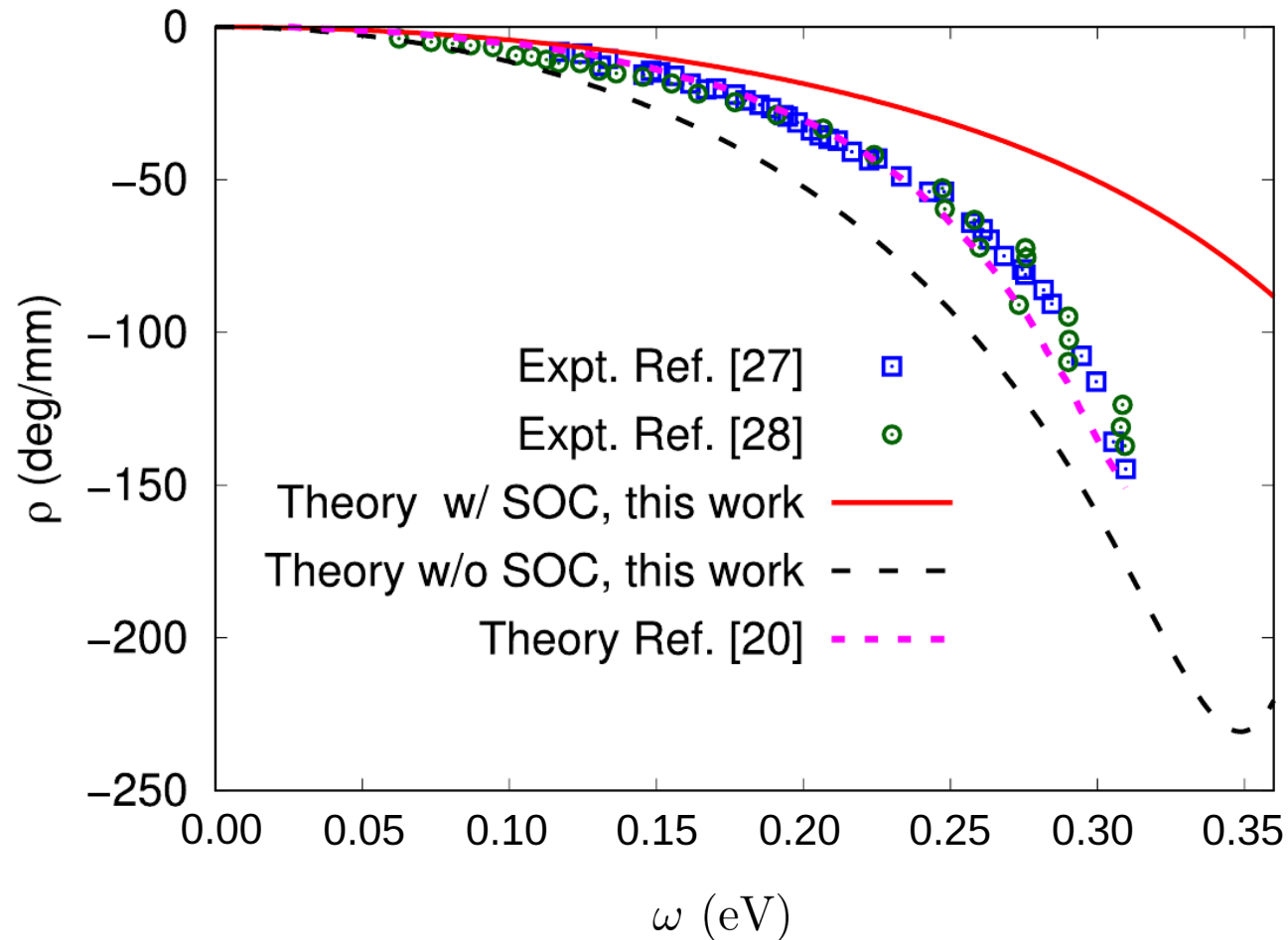


Rotatory power: $\rho = \frac{\beta}{d}$



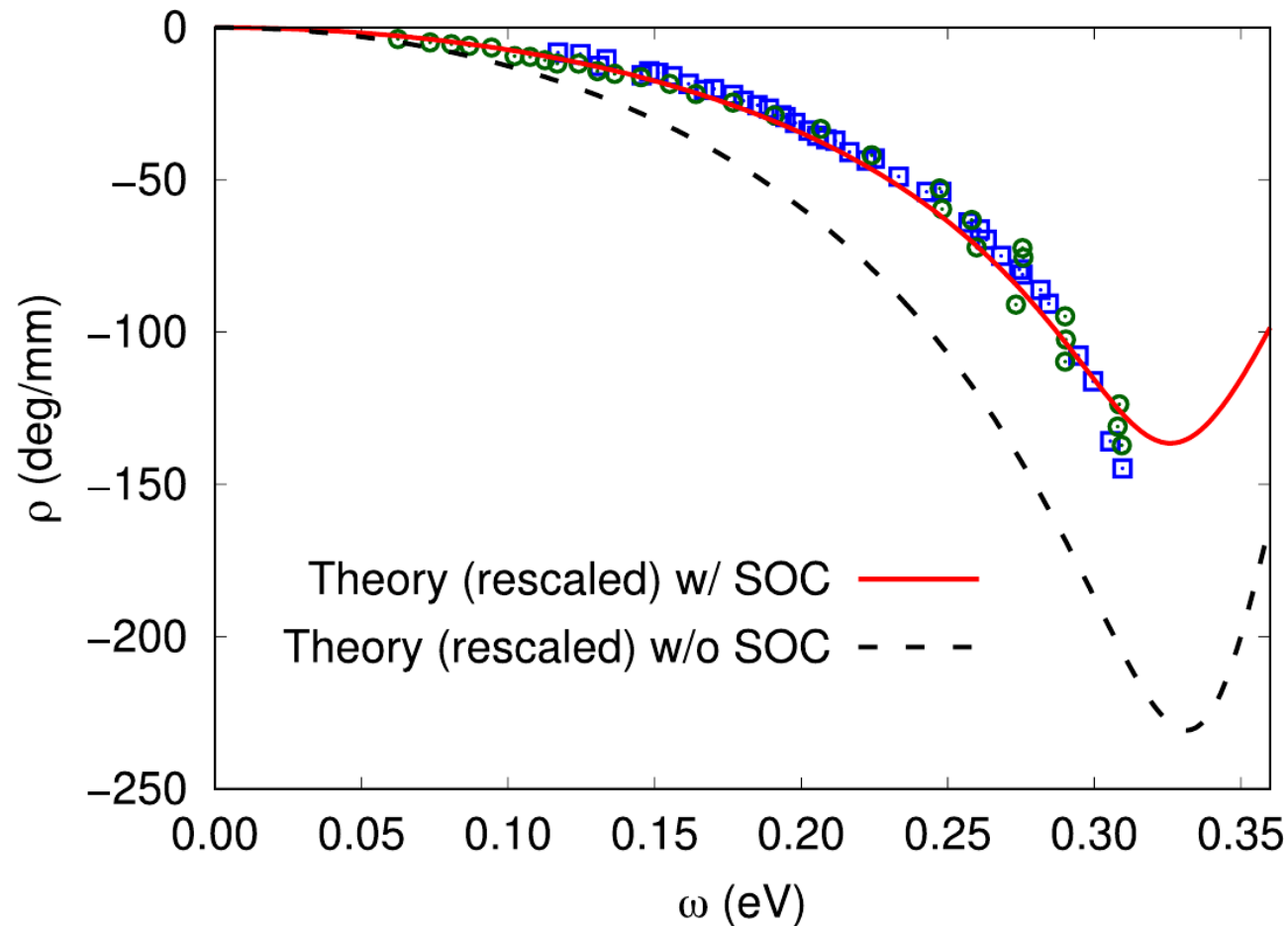
Trigonal tellurium

Rotatory power of trigonal tellurium



Raw calculated data compared to experiment and to a previous theoretical calculation [20].

Rotatory power of trigonal tellurium



With a rescaled frequency axis for the theoretical results, to align *ab initio* energy gap with the experimental one.

Summary

- Wannier-based *ab initio* scheme for optical spatial dispersion in crystals
- Implemented in WannierBerri (wannier-berri.org)
- First application: optical activity

Outlook

- Directional dichroism in AFM insulators (**Cr₂O₃**)
- Low- ω intraband responses in broken-symmetry conductors
 - Edelstein effect (orbital+spin): intraband optical activity in chiral conductors (**p-doped Te**)
 - Quantum metric dipole: intraband directional dichroism in AFM conductors (**CuMnAs**)